CE 310

Shape Matters — Reading Distributions

Week 2 · Session 1 · Histograms, Skewness, Box Plots, and Percentiles

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Part 1

Why Shape Matters

Four Reports Land on Your Desk

Four teams each tracked a quantity for a year, reported as **actual ÷ expected**:

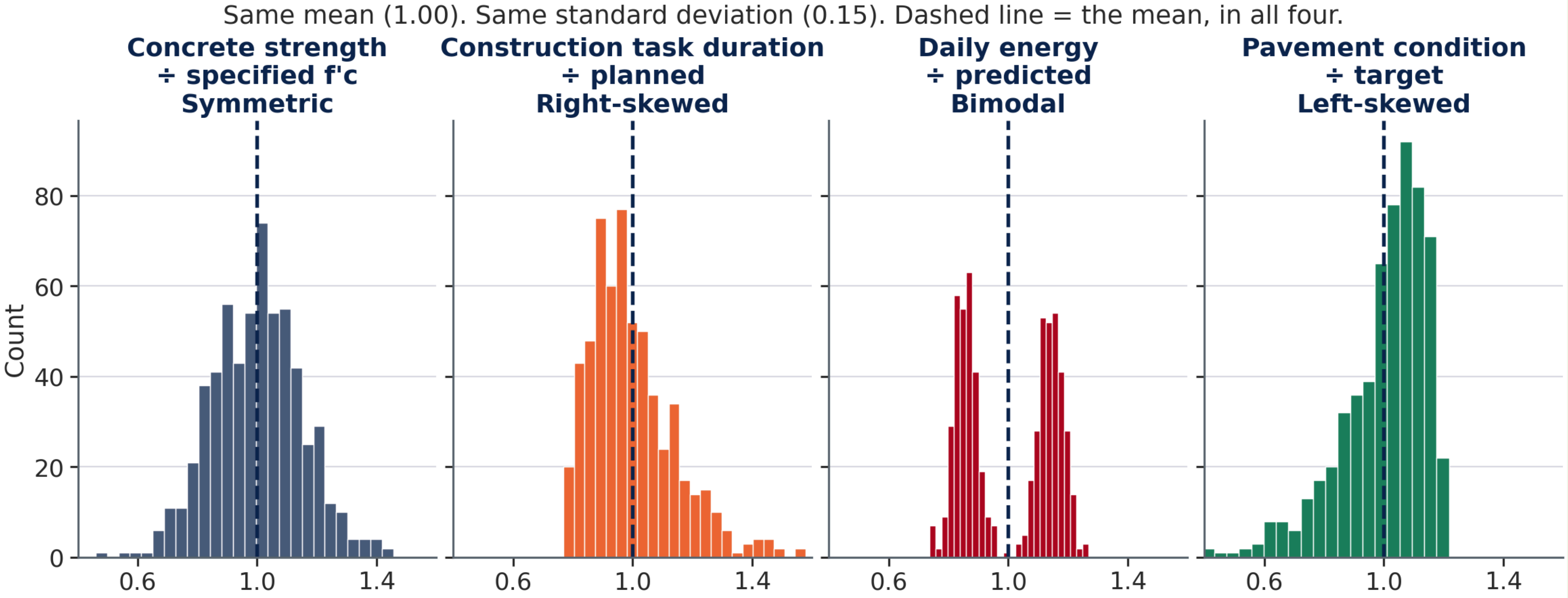
	Concrete strength ÷ f'c	Construction task duration ÷ planned	Daily energy ÷ predicted	Pavement condition ÷ target
Observations	600	600	600	600
Mean	1.00	1.00	1.00	1.00
Standard deviation	0.15	0.15	0.15	0.15

A *construction task* is one line on the project schedule — pour footings, hang ductwork, pave a lift. The schedule gives it a planned duration; the ratio is how long it actually took.

Are these four processes behaving the same way? One will blow up your schedule, one will make your energy model useless, one is genuinely fine.

> ⚠️ Commit to an answer now. You have the two numbers engineers report most often — and they are identical.

✅ Same Two Numbers. Four Different Shapes.



💡 Two numbers cannot carry a **shape** — and the decision lives in the shape.

What Each Shape Changes

Shape	What it means for your decision
Symmetric	The mean *is* the typical batch. Design margins behave the way the textbook says.
Right-skewed	Most tasks land near the estimate; a few run far over. **Median 0.96** — planning to the mean already assumes you are unlucky.
Bimodal	Two regimes — occupied and unoccupied days. **The mean, 1.00, essentially never occurs.** It describes neither.
Left-skewed	Most segments sit near target, with a tail of degraded ones. **Median 1.04** — the average makes the network look worse than most of it is.

⚠️ All four medians differ — 0.96, 1.00, 1.01, 1.04 — and nobody reported the median. The shape was visible in the data the whole time.

Your Turn — Predict the Shape

Before we look at the data: based on what you know about Tucson's climate, predict the shape of each distribution.

Variable	Your prediction: Left / Symmetric / Right?
Annual temperature (hourly, full year)	???
Wind speed (hourly, full year)	???
Relative humidity (hourly, full year)	???

> Think about it for 90 seconds. We'll reveal the answer after the histogram slides.

✅ Predict the Shape — Answers

Tucson 2023 hourly data

Variable	Actual shape	Physical reason
Temperature	Nearly **symmetric**	Cold winter left tail ≈ hot summer right tail
Wind speed	**Right-skewed**	Speed cannot be negative; calm hours dominate, rare storms pull right
Relative humidity	**Right-skewed**	Desert is mostly dry; monsoon season pulls the right tail

> 📘 **Right-skewed** (positive skew): the distribution has a long tail on the right side. A few unusually high values pull the mean above the median. Most observations cluster at lower values; rare high extremes are possible.

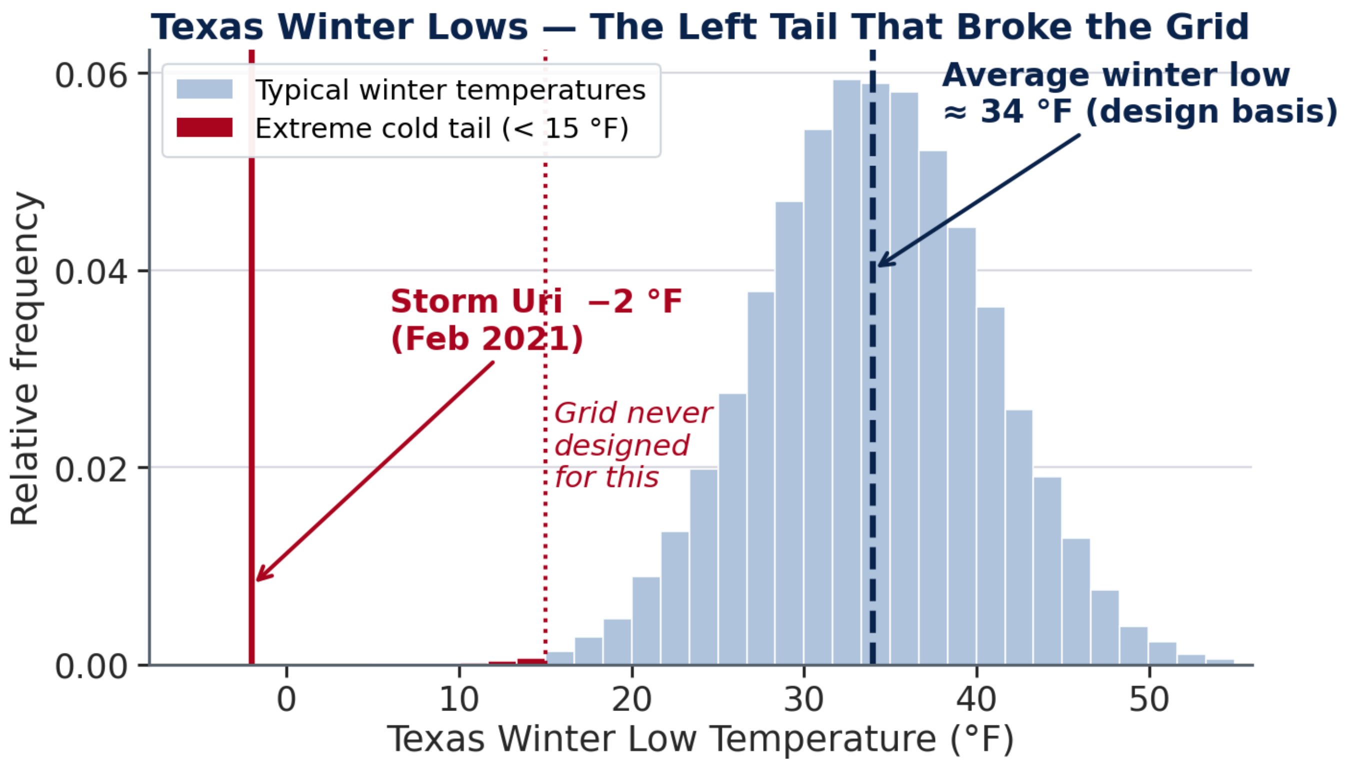
That Left Tail, in a Real System

Texas winter temperatures are left-skewed — mild most years, with rare extreme cold in the tail.

ERCOT — the Electric Reliability Council of Texas, operator of the state's self-contained grid — sized winter generating capacity against **average** winter conditions.

February 2021 · Storm Uri: –2°F, 40+ plants offline within 48 hours, 246 deaths.

The mean was not wrong. It was **not the question.**



Why this becomes a blackout. Extreme cold drives demand *up* — electric heating — in the same hours it drives supply *down*: frozen wellheads, gas lines and un-winterized plants. Capacity sized for the average has no margin for both at once, so the operator cuts power to whole neighborhoods on purpose, to stop the rest of the grid going down with them. ERCOT has said it came within minutes of exactly that — a collapse the state would have restarted plant by plant, over weeks.

Week 1 Said "Not the Average." Which Number, Then?

Week 1 established the rule — a system sized for the average fails exactly when it is stressed most — and stopped there. It never named the number to use instead.

Design decision	Week 1: not this	Week 2: this
Size HVAC cooling system	Annual average temp (72°F)	99th percentile temp **(107.5°F)**
Design wind load for signage	Average wind speed (7.2 mph)	95th percentile wind **(17.2 mph)**
Size dehumidification coil	Annual average RH (32.7%)	90th percentile RH **(64.1%)**
Plan heating capacity (TX)	Assume symmetric winter temps	Recognize the **left tail**

> 💡 Every entry in the right-hand column is a **position in the distribution**, not a summary of it. No mean and standard deviation will give you one — which is why the rest of this lecture is about shape.

Part 2

The Histogram

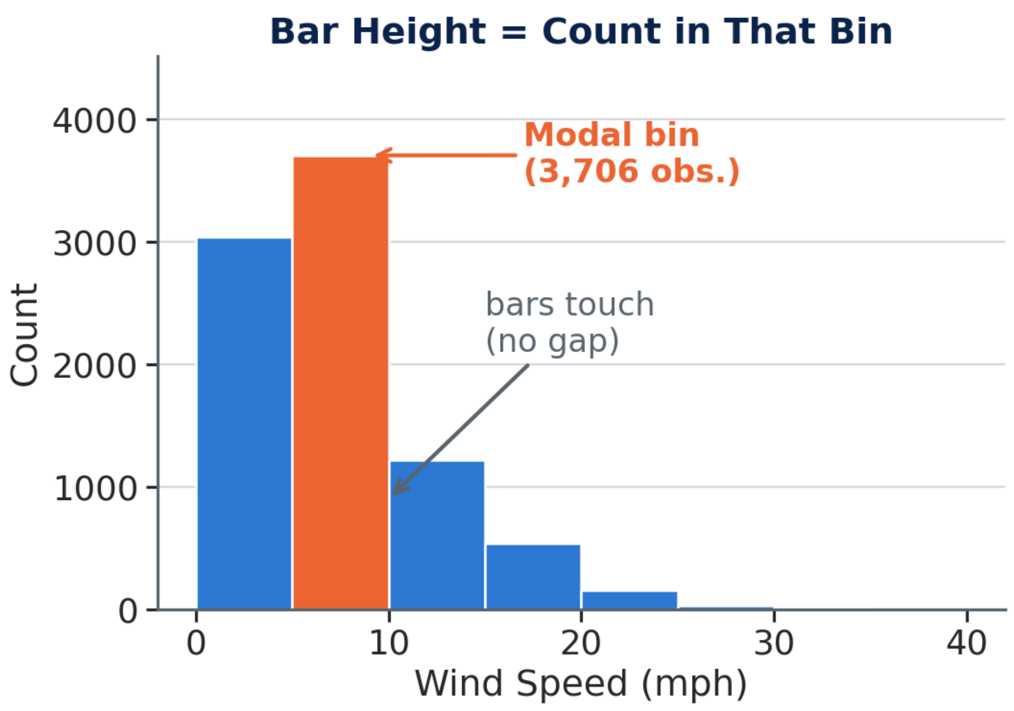
What Is a Histogram?

📘 **Definition:** A histogram divides a **numeric** variable into equal-width **bins** and counts how many observations fall in each bin. The height of each bar = count of observations in that bin.

Key properties:

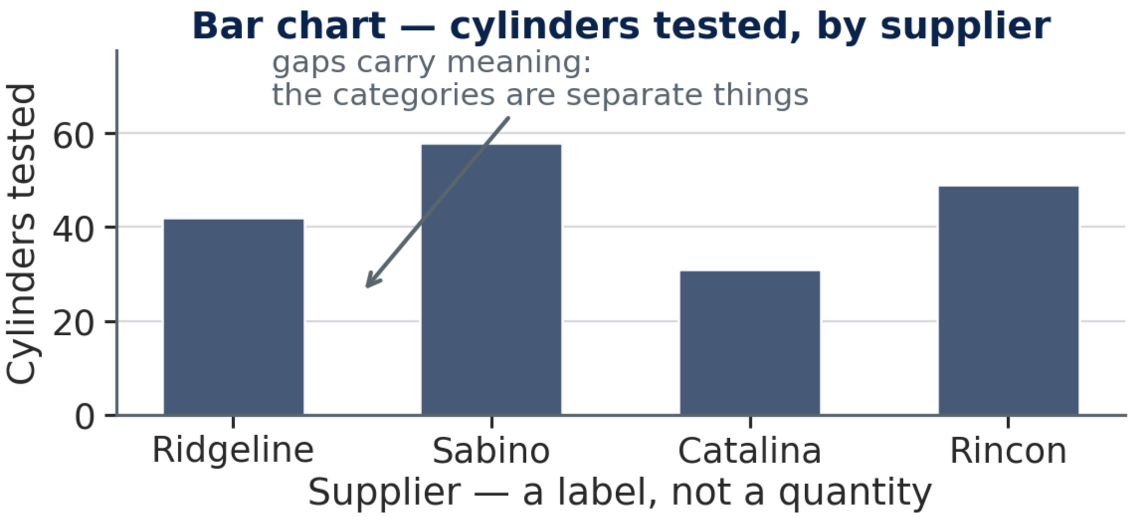
- Bins are **equal width**; bars **touch** — no gap (unlike a bar chart)
- x-axis: any **numeric** scale — continuous (°F) or whole counts (trucks per hour); y-axis: **count**
- Reveals **shape**, **clusters**, and **outliers**
- **Bin width is a choice you make**, not a property of the data

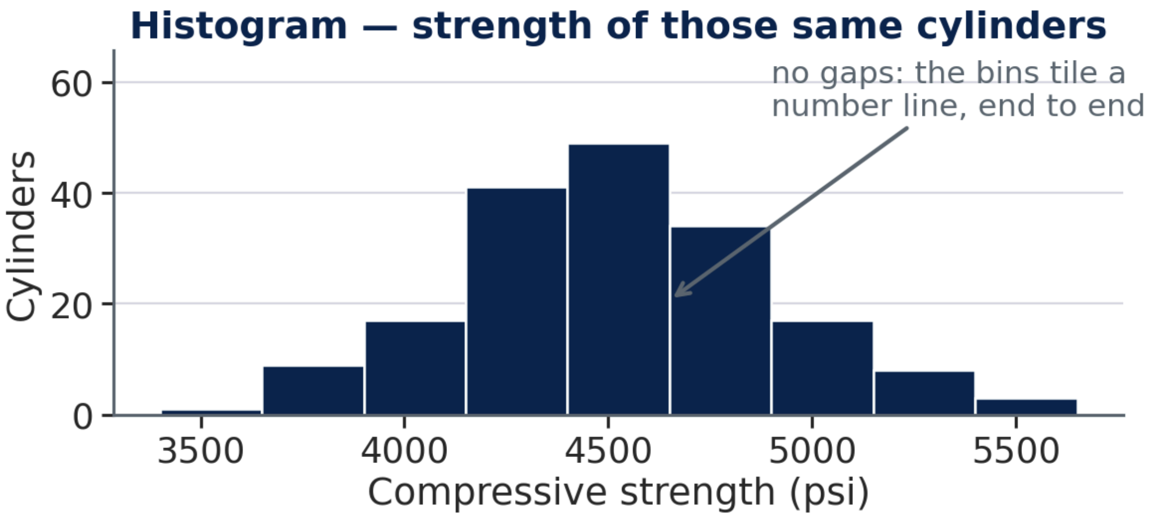
⚠️ Too few bins smooth a second peak away; too many turn sampling noise into apparent structure. Excel's automatic 8–10 bins is a starting point, not an answer — check more than one bin width before describing a shape.



Bar Chart or Histogram? Look at the x-Axis

Both draw rectangles. What differs is **what the horizontal axis carries.**





	Bar chart	Histogram
x-axis holds	Categories — supplier, material, district	A number line — psi, °F, mph
Bars	Separated; the gap says these are distinct things	Touching; the bins tile the axis, nothing in between

💡 **The test: could you reorder the bars?** On the left, yes — nothing is lost. On the right, no; a bar's position *is* its value, which is why only a histogram has a shape.

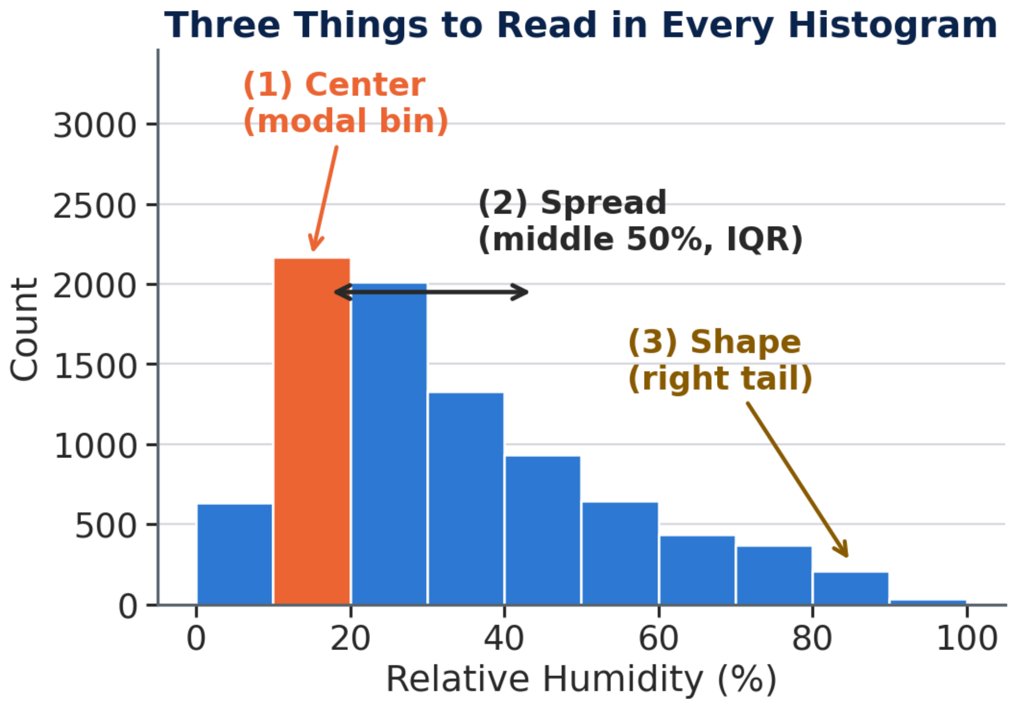
Reading a Histogram — What to Look For

(1) Center — where does the bulk of the data sit? Is the peak near the mean, or off to one side?

(2) Spread — how wide is the distribution? Values clustered tightly or spread over a wide range?

(3) Shape — symmetric? One tail longer than the other? One peak or two?

> ⚠️ A symmetric distribution and a right-skewed one can share the same mean and SD — shape reveals what those numbers hide.



One Peak or Two? — Bimodal Distributions

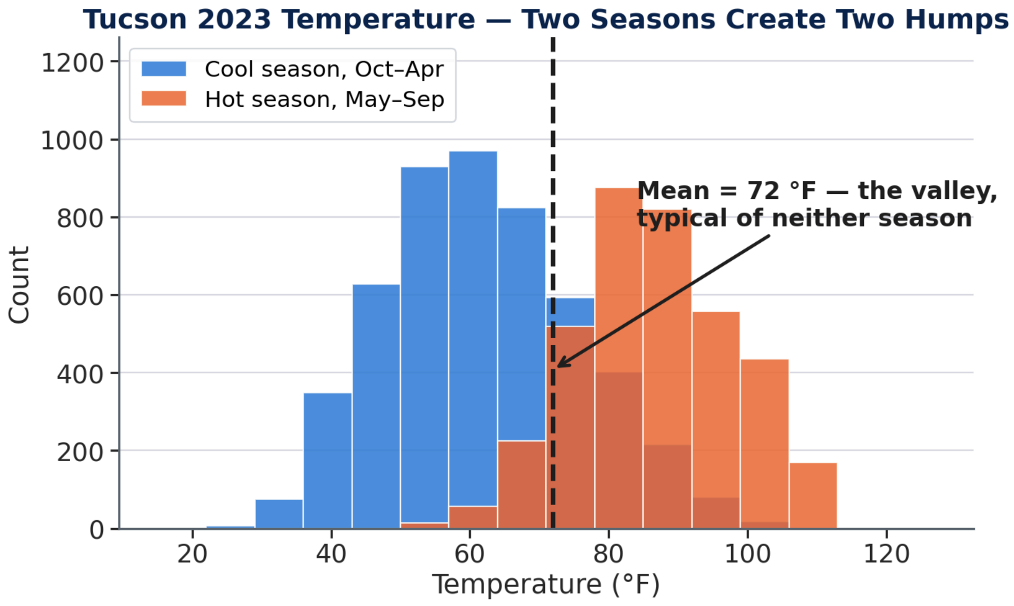
Most distributions have **one peak** (unimodal). But mixing two subpopulations creates **two peaks** (bimodal):

Common causes:

- Two distinct seasons or occupancy modes
- Equipment with two operating states (on vs. standby)
- A process that changed mid-measurement

Warning signs:

- Two humps with a valley between them
- Mean falls **in the valley** — typical of neither group
- Skewness ≈ 0 can mislead — a symmetric bimodal still has two typical values



⚠️ Two humps signal a mixed population — split by subgroup before computing any statistics.

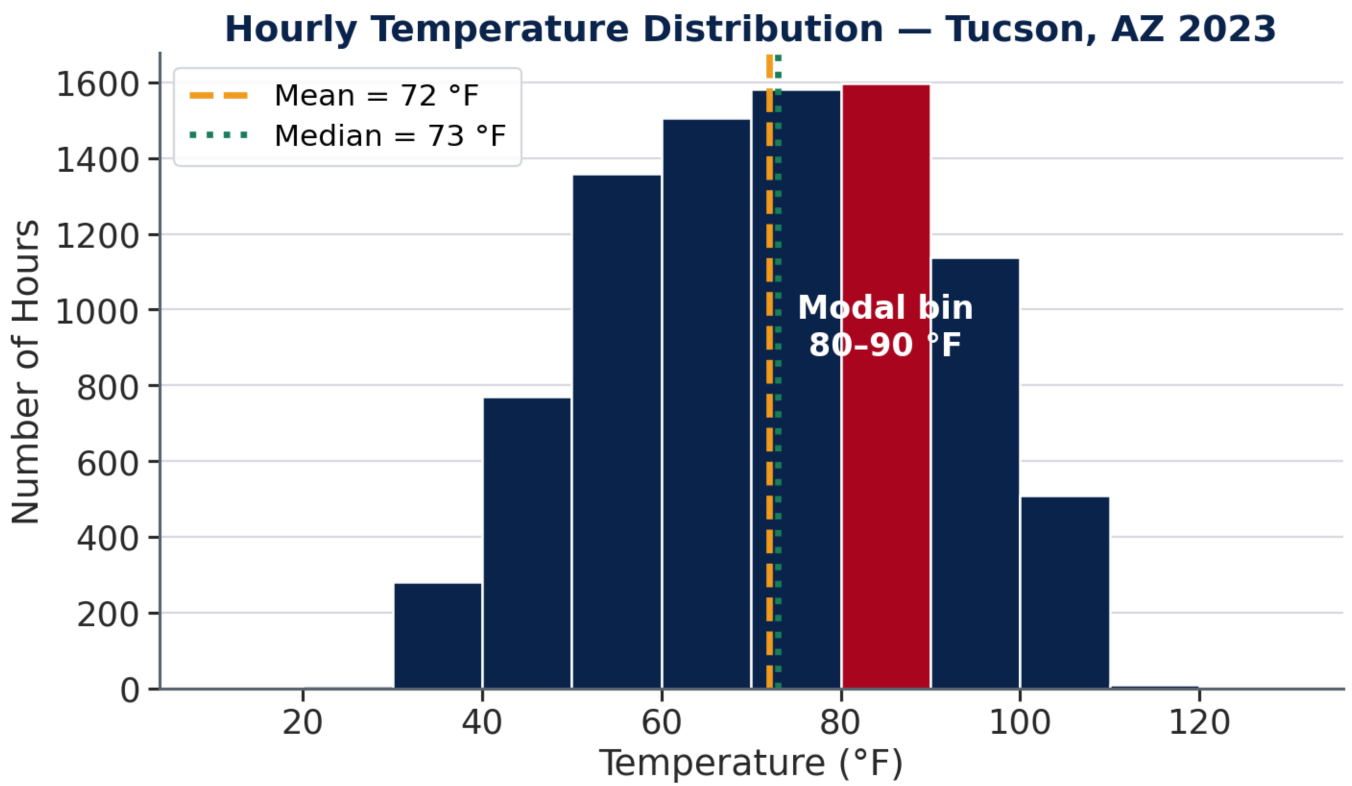
Tucson Temperature — What Does the Histogram Show?

- **Modal bin: 80–90°F** — most common temperature range
- **Mean (72°F) ≠ Mode (85°F)** — they differ by 13°F
- **Plateau shape**, not a bell curve

Why the plateau?

Two overlapping seasons:

- Cool season (Dec–Mar): 40–65°F
- Hot season (Jun–Sep): 85–110°F



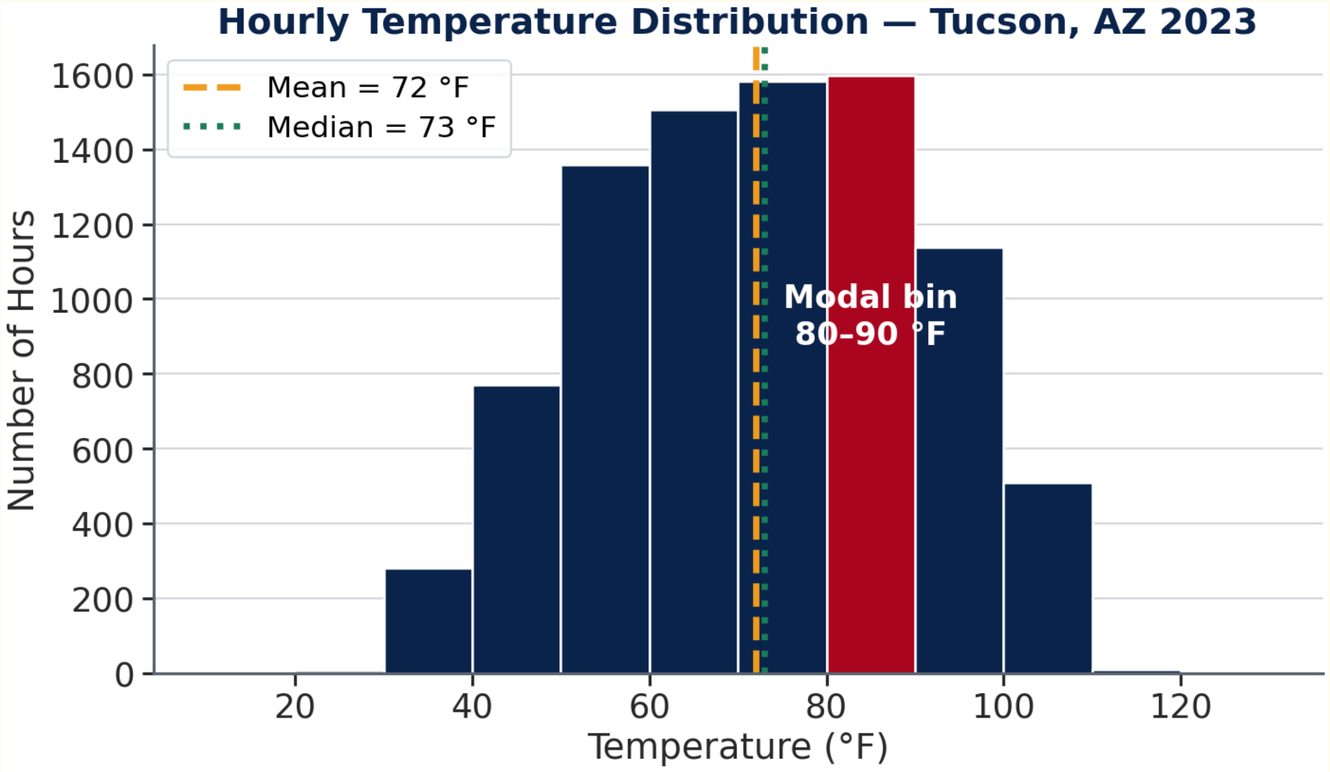
Look at This Histogram

Based on what you see — is this distribution:

A) Left-skewed B) Symmetric C) Right-skewed

Clues to check:

- Where is the **peak** (tallest bar)?
- Is one **tail longer** than the other?
- Is the mean **(gold dashed)** to the left or right of the median **(white dotted)**?



✅ Answer — Nearly Symmetric (skewness = –0.05)

The mean (72°F) and median (73°F) differ by only 1°F — a near-perfect match, the hallmark of symmetry.

Why it looks left-skewed but isn't:

- Modal bin (80–90°F) is above the mean
- This looks like left skew — but the **cold winter tail** (20–50°F) nearly cancels the **hot summer tail** (90–110°F)
- Net result: almost symmetric

What would be surprising about wind?

- Wind speed **cannot be negative**
- Most hours: calm (0–10 mph)
- Occasional storms: 20–34 mph
- This forces a **right tail** → right-skewed

> 💡 A skewness coefficient near zero (–0.05) confirms what the mean ≈ median already suggested.

Part 3

Measuring Shape — Skewness

What Skewness Measures

📘 **Skewness** quantifies the asymmetry of a distribution. It tells you whether one tail is longer than the other — and by how much.

$$g = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^3$$

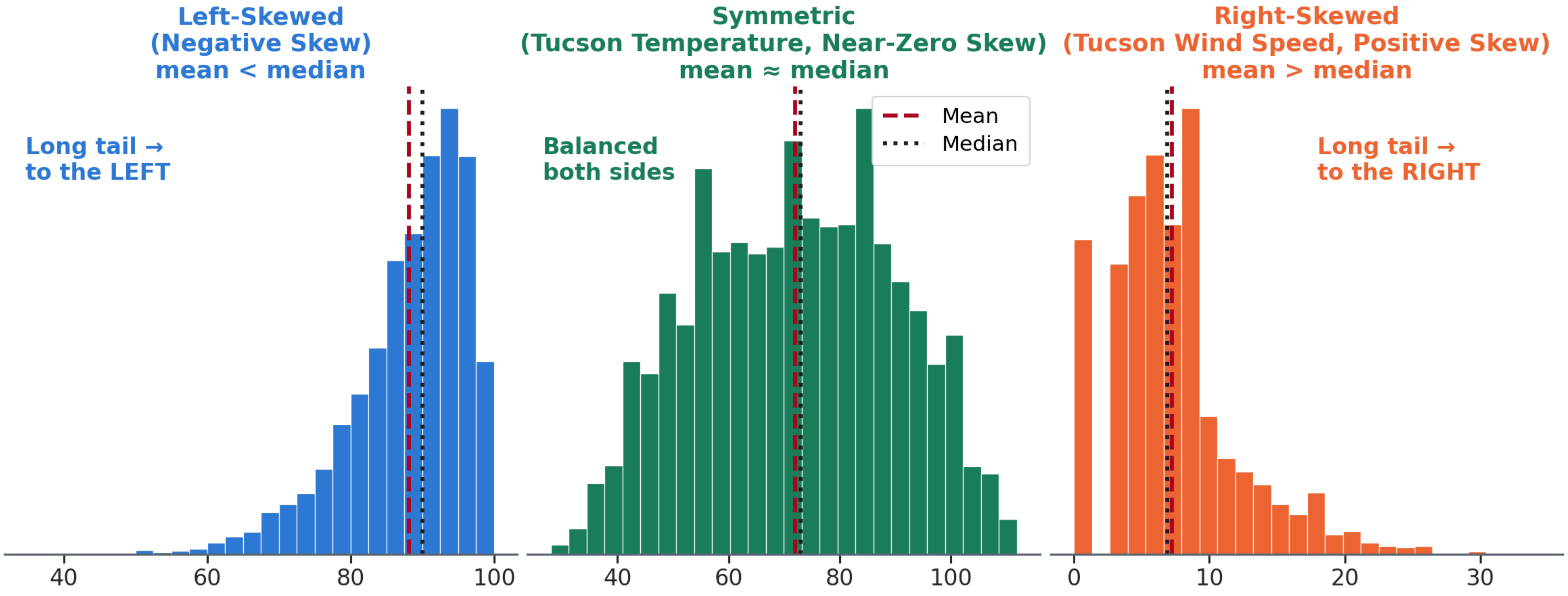
g = the skewness value · x_i = each value · $\bar{x}$ = sample mean · s = sample standard deviation

Why cubed? The odd power keeps the sign — a long left tail pulls g negative. Squaring would erase direction; that is variance. Dividing by s first makes g dimensionless, so mph and psi are comparable.

Excel applies a small sample-size correction to this formula. At $n = 8{,}760$ the difference is under 0.1% — ignore it.

Value of g	Meaning
Near 0 (–0.5 to +0.5)	Symmetric — mean ≈ median
Negative (< –0.5)	Left-skewed — long left tail, mean **<** median
Positive (> +0.5)	Right-skewed — long right tail, mean **>** median

The Three Shapes



Left Skew vs. Right Skew — What Pulls the Mean

Left-skewed (negative)

- Tail extends to the LEFT
- A few very LOW values pull the mean DOWN
- Mean < Median
- Example: exam scores where most students do well but a few fail badly

Engineering example: Strength of materials — most samples are at or above spec, but rare defects create a low-value left tail

Right-skewed (positive)

- Tail extends to the RIGHT
- A few very HIGH values pull the mean UP
- Mean > Median
- Example: income — most people earn modest amounts, a few billionaires pull the average up

Engineering example: Construction task durations — the opener's right-skewed panel, **median 0.96 vs. mean 1.00.** Wind, rainfall, and flood flows share the shape

The Three Tucson Variables — Revealed

Variable	Mean	Median	g	Shape
Temperature (°F)	72	73	**–0.05**	Nearly symmetric
Wind speed (mph)	7.2	6.9	**+1.07**	Right-skewed
Relative humidity (%)	32.7	27.6	**+0.92**	Right-skewed

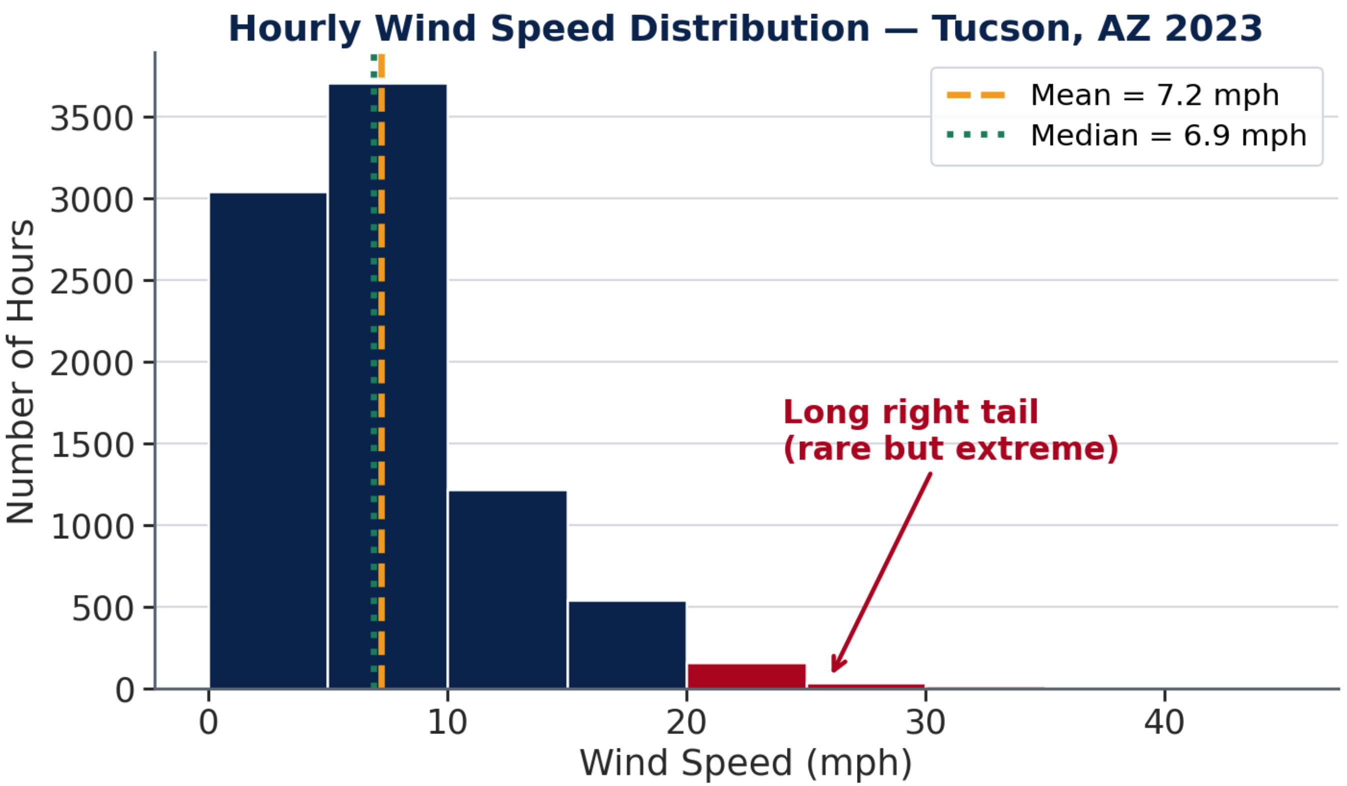
> 💡 **Wind and humidity are both right-skewed** — most hours are calm/dry, but rare monsoon and storm events create the high-value right tails. This is why using the *average* wind or humidity for design grossly underestimates the worst conditions.

Wind Speed — The Right Skew in Action

Wind speed statistics:

- Mean: 7.2 mph · Median: 6.9 mph
- Skewness g: **+1.07** (right-skewed)
- P95: **17.2 mph** (2.4× the mean)
- **P95 wind pressure = 5.7× average** (wind pressure $\propto V^2$)

Why right-skewed? Speed cannot be negative — bounded at zero. Most hours are calm; rare storms form the right tail.



Quick Check — Classify the Shape

Three datasets, three skewness values. Match each to the variable — and explain the physical reason for that shape.

Dataset	g	Your call: Left / Symmetric / Right?
Dataset X	−1.4	???
Dataset Y	+0.1	???
Dataset Z	+2.3	???

Which variable is which?

- **A)** Concrete cylinder compressive strength (100 lab-controlled samples)
- **B)** Bore diameter of 500 machined steel bearings from one production run
- **C)** Daily peak water demand at a city treatment plant

✅ Quick Check — Answers

Dataset	g	Variable	Physical reason
X	−1.4	Concrete strength	Most samples meet or exceed spec; rare defective batches pull the left tail down
Y	+0.1	Bearing bore diameter	A controlled machining process centers on the target; random tool and thermal variation scatters evenly both ways → symmetric
Z	+2.3	Daily water demand	Most days near baseline; rare heat waves or large events create a long right tail

💡 **Physical reasoning predicts shape before you plot.** Ask: Is there a zero floor? (→ right skew.) Is there a quality target most samples exceed? (→ left skew.) Is the process tightly controlled? (→ symmetric.)

Part 4

The Box Plot

The Five-Number Summary

In Week 1 you computed Q1, Q3, and IQR for Tucson temperatures. Those five numbers — Minimum, Q1, Median, Q3, Maximum — are the **five-number summary**: the foundation of every box plot.

Statistic	Value	Meaning
Minimum	28°F	Coldest single hour of the year
Q1	58°F	25% of hours are colder than this
Median	73°F	Half of hours are above, half below
Q3	86°F	75% of hours are colder than this
Maximum	112°F	Hottest single hour of the year
IQR	**28°F**	Spread of the middle 50% — Q3 – Q1

💡 These map directly onto the box plot: the box spans Q1 to Q3, the line inside is the median, and the whiskers extend toward the Min and Max.

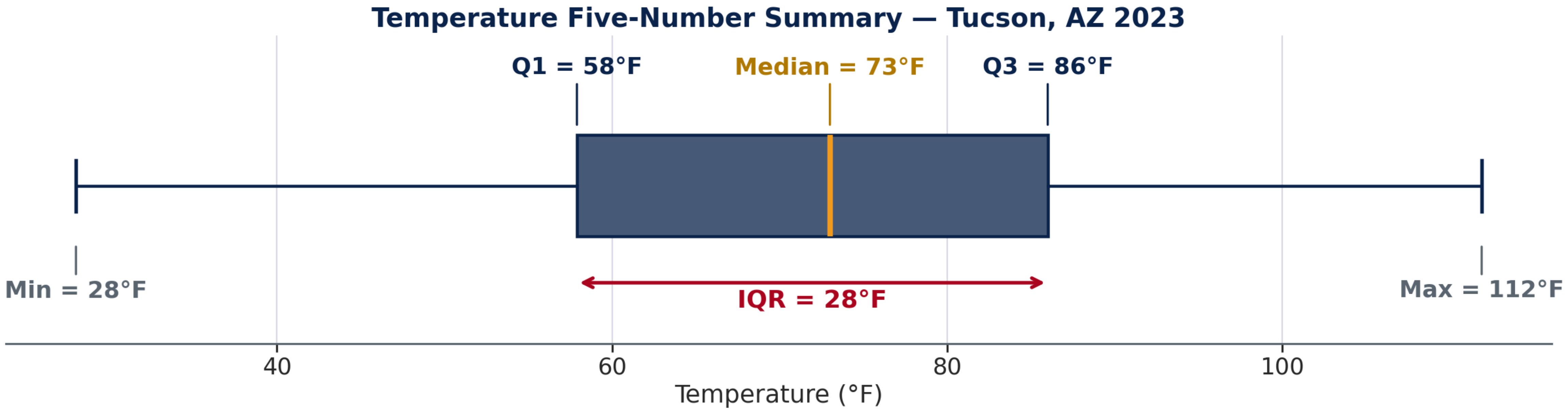
Reading a Box Plot

The box spans Q1 to Q3 — the middle 50% (IQR)

The line inside is the median (not the mean)

The whiskers extend to the most extreme values within 1.5 × IQR

Dots beyond whiskers are outliers



Where the Whiskers Stop — The 1.5 × IQR Rule

A whisker does not run to the most extreme value it can find. It stops at a computed boundary called a **fence**, and only points beyond a fence are drawn as outlier dots.

The rule — four steps

1. **IQR** = Q3 – Q1
2. Multiply: **1.5 × IQR**
3. **Lower fence** = Q1 – 1.5 × IQR
   Upper fence = Q3 + 1.5 × IQR
4. Anything beyond a fence is an **outlier**

The 1.5 is a convention, not a law — some fields use 2 × or 3 × IQR.

Worked on Tucson temperature

- IQR = 86 – 58 = **28°F**
- 1.5 × 28 = **42°F**
- Lower fence = 58 – 42 = **16°F**
- Upper fence = 86 + 42 = **128°F**
- Coldest hour 28°F, hottest 112°F — both **inside** the fences

Outliers flagged: none.

💡 The identical rule on wind speed flags **581 hours** — 6.7% of the year, all at the high end. Same file, same rule, opposite verdict: temperature is near-symmetric, wind is right-skewed, and the tail is where the outliers live.

Why 1.5? Where the Number Came From

John Tukey, who introduced the box plot in 1977, chose the multiplier so that a well-behaved symmetric process flags **almost nothing** — which is what makes anything it *does* flag worth a look.

In a symmetric, bell-shaped distribution Q1 and Q3 sit 0.67 SD either side of the center, so IQR = 1.35 SD. That fixes where each candidate fence lands:

Multiplier	Fence sits at	Share of a symmetric process flagged
1.0 × IQR	2.02 SD	4.3% — one point in 23. Too noisy to mean anything
1.5 × IQR	**2.70 SD**	**0.7% — one point in 143.** Rare, but not impossible
3.0 × IQR	4.72 SD	0.0002% — Tukey's "far out" fence, for the extreme only

💡 1.5 is a **calibration**, not a law. It encodes an answer to "how unusual is unusual enough to look at?" — roughly one point in 150. A right-skewed variable like wind trips it far more often, and that is the finding, not a fault in the rule.

IQR vs. Standard Deviation — What's the Difference?

Standard Deviation (SD)

- Measures spread of **all** values
- Heavily influenced by **outliers** and **extreme values**
- Assumes roughly symmetric distribution for easy interpretation
- Tucson temperature SD: **17.9°F**

Interquartile Range (IQR)

- Measures spread of the **middle 50%**
- **Outlier-resistant** — extreme values don't affect it
- Works well for skewed distributions
- Tucson temperature IQR: **28°F**

> 💡 For right-skewed data (wind, humidity, rainfall), IQR is a more reliable spread measure than SD because it ignores the long tail. For symmetric data (temperature here), SD and IQR give complementary information.

Quick Poll — IQR vs. SD

Two cities have the same annual average temperature (72°F) and the same standard deviation (18°F). But City A has an IQR of 28°F and City B has an IQR of 12°F.

What does this tell you about the two cities?

- A) They have identical climates — same mean and SD means same distribution
- B) City A has a more uniform climate across the year; City B has more concentration near the average
- C) City B has a more concentrated middle 50% — tighter "typical" temperature band; City A has wider seasonal swings
- D) Not enough information to say anything

✅ Quick Poll — The Answer Is C

City B has the tighter typical temperature band

	City A	City B
Annual mean	72°F	72°F
Standard deviation	18°F	18°F
IQR	**28°F**	**12°F**
Climate character	Wide seasonal swings	Narrow "typical" band

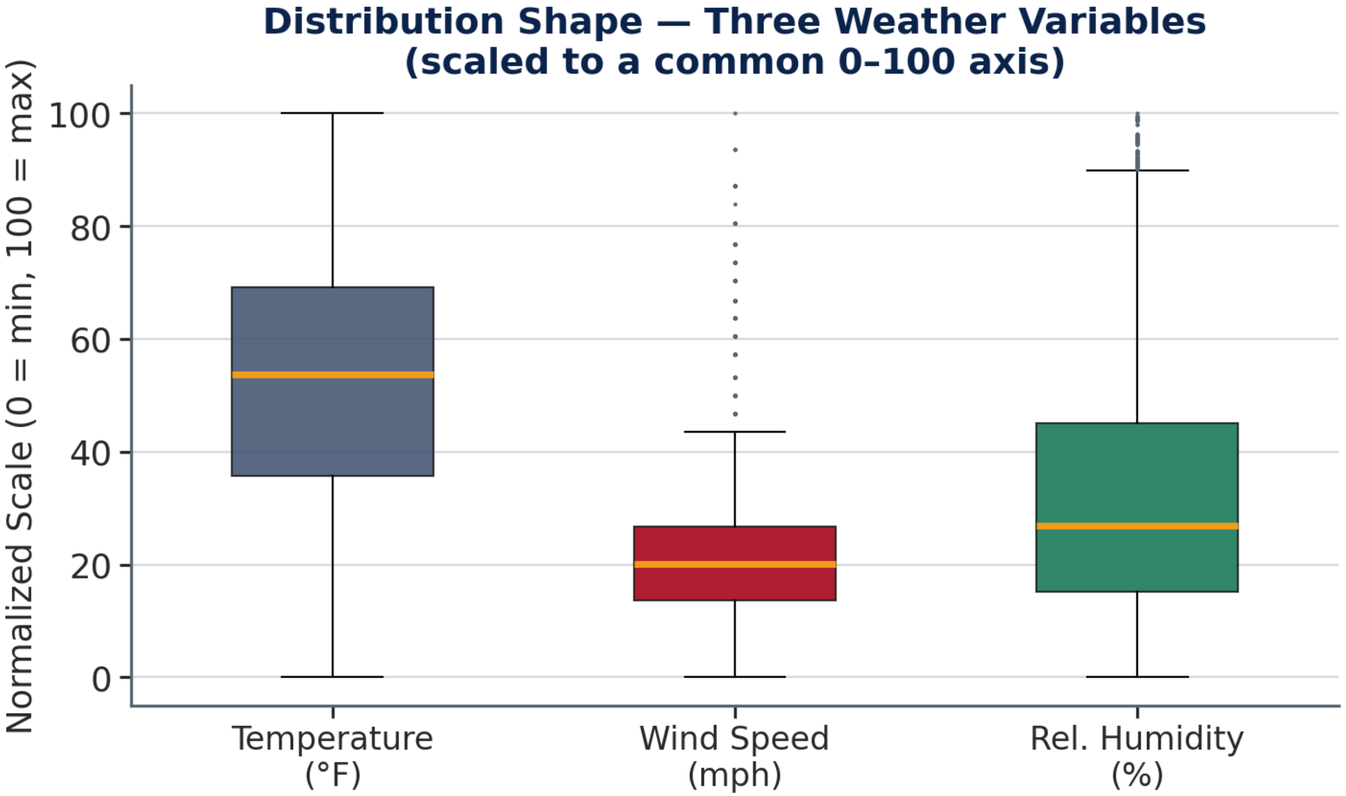
Why same SD but different IQR? City A's large IQR reflects broad seasonal swings (cold January, hot July); City B's hours cluster near the mean, offset by occasional extremes that push the SD up.

> 💡 SD reacts to extremes; IQR reflects the middle 50%.

Box Plots Side by Side — Three Variables

All three scaled to a common 0–100 axis:

- Temperature: **wide box, median centered** — IQR = 28°F
- Wind: **narrow box, long upper whisker + outliers** — IQR = 4.5 mph
- Humidity: **wide box, median low in it** — IQR ≈ 30 pts



> 💡 Right skew reads two ways here: a squeezed box with a long whisker (wind), or a wide box with the median pushed toward its bottom edge (humidity).

Part 5

Percentiles and Design Thresholds

Design Percentiles — The Full Spectrum

In Week 1 you found the ASHRAE 1% design temperature — the 88th-largest reading, the 99th percentile by rank. Here is what the full spectrum looks like for Tucson:

Tucson 2023:

Percentile	Temperature	Meaning
25th (Q1)	58°F	25% of hours are below 58°F
50th (Median)	73°F	Half the year is below 73°F
75th (Q3)	86°F	75% of hours are below 86°F
90th	96.1°F	799 hours (10%) are above this
95th	**100.0°F**	**5% exceedance** — 429 hours above this
99th	**107.5°F**	**ASHRAE 1% cooling design condition** — 88 hours above

💡 Week 1 got **108.0°F** for this same condition. Both are right: `LARGE` returns the 88th-largest reading you actually have; `PERCENTILE` interpolates between the 88th and 89th (108.0 and 107.1). Same 88 hours above it either way.

Design Percentiles — Why Not 100%?

Engineering design is always a tradeoff between **cost** and **risk**:

100th percentile (absolute maximum)

- Guarantees performance under every observed condition
- Extremely expensive to design for
- The maximum may be a freak event never to repeat

90th–99th percentile

- Handles the vast majority of conditions
- Much more economical
- Residual risk is **known and accepted**

Standard engineering practice:

- HVAC cooling loads: **99th or 98th** percentile dry-bulb — the ASHRAE 1% and 2% conditions
- Structural wind loads: **50-year return period** (~98th annual percentile)

Precision Check — Percentile ≠ Probability

⚠️ **P90 does not mean "90% probability."** It means 90% of observed values fall *below* this point — a rank in your data, not a forecast.

Formal statement: The Pth percentile is the value below which P% of all observations fall; (100 – P)% are above it.

Statement	Plain English
P90 of wind speed = 13.9 mph	90% of all hours are calmer than 13.9 mph
P95 of temperature = 100°F	5% of all hours are hotter than 100°F
P99 of temperature = 107.5°F	1% of all hours exceed 107.5°F — the ASHRAE 1% cooling design condition

The key insight: A percentile is a **location** in the distribution — not a probability, not a maximum.

From Definition to Number — The Logic Behind a Percentile

Example: Find the 90th percentile of Tucson wind speed (8,700 valid hourly readings)

Step 1 — How many values fall below P90?

> 8,700 × 0.90 = **7,830 values**

Step 2 — What is the 7,830th-lowest value?

> Sort low to high → position 7,830 → **P90 = 13.9 mph**

Step 3 — Verify the interpretation:

> 7,743 hours (89.0%) are strictly below 13.9 mph
> 219 more hours read *exactly* 13.9 — the 7,830th lands inside that block ✓
> 738 hours lie above it — the windiest 10% of the record ✓

> 💡 The tool sorts and counts. You supply the judgment about *which* percentile the design requires.

Exceedance Probability — The Language of Design Standards

Exceedance probability = the fraction of time a threshold is exceeded.

$$P(\text{exceedance}) = 1 - \frac{p}{100} \qquad (p = \text{design percentile, e.g. } p = 95 \Rightarrow 5\% \text{ exceedance})$$

Design percentile	Wind speed	Exceedance	Hours above, in this record
P90	13.9 mph	10%	738 hr
P95	**17.2 mph**	**5%**	**342 hr**
P99	23.0 mph	1%	66 hr

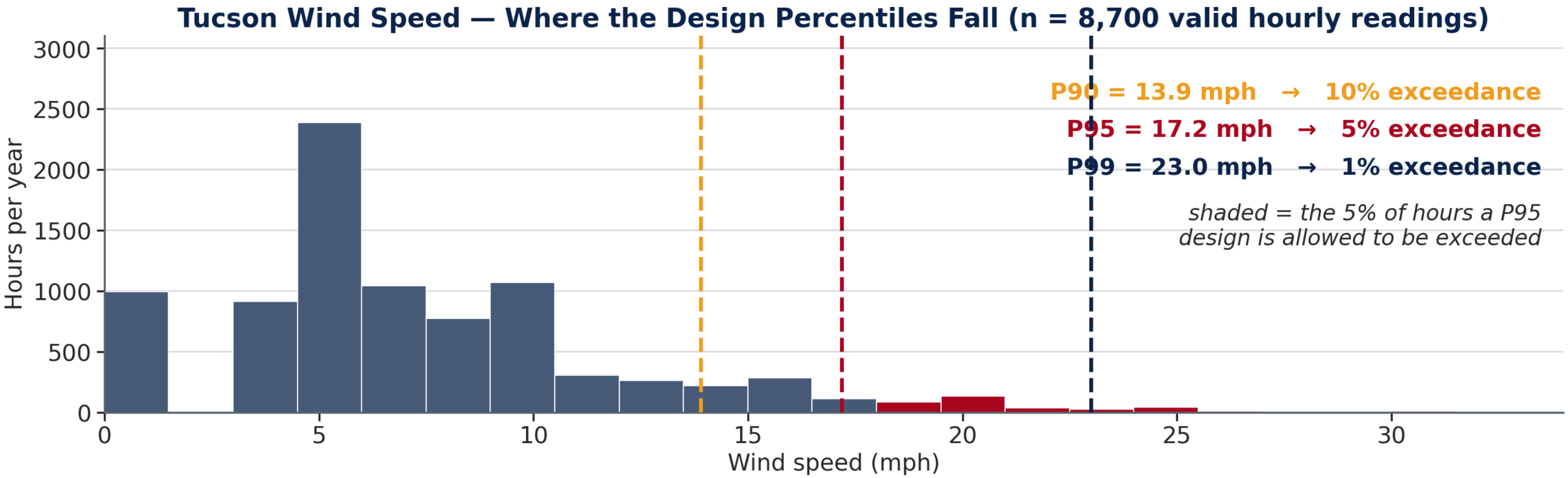
Why engineers use this language:

- ASHRAE: "0.4% cooling condition" = P99.6 temperature

> 💡 Design standards are written in exceedance language. The hour counts are close to 10/5/1% of a year, not equal to it — real records have gaps and repeated readings.

All Three Percentiles, On One Distribution

The same wind data from Part 3 — now with the design thresholds drawn on it.



Quick Check — Design Threshold Decision

A Tucson building engineer sized a dehumidification coil to handle humidity at the **90th percentile (RH = 64.1%)**.

Two questions:

(a) The dataset has 8,760 hours. How many hours per year will outdoor humidity **exceed** the coil's design capacity?

(b) The building owner asks: "*Will the dehumidification system always keep us comfortable?*" How do you answer — honestly and professionally?

✅ Design Threshold — Answers

(a) Hours above design capacity per year:

> 10% × 8,760 = **876 hours/year** (≈ 36 days — primarily July–August monsoon season)

(b) The honest professional answer:

> "Not always. The system is designed to the ASHRAE 62.1 90th-percentile standard, meaning outdoor humidity will exceed the coil's capacity roughly 876 hours per year. This is the industry-accepted level of risk. A system designed for the 99th percentile would handle 99% of hours but would cost substantially more to build and operate. The choice of design percentile is a cost-risk tradeoff made explicitly at design time."

The principle:

Every design percentile is a **decision to accept a known exceedance frequency**.

Part 6

Putting It All Together

A Complete Design Workflow — Four Tools, One Decision

Scenario: A civil engineer must specify the design wind speed for a Tucson billboard structural attachment. City code requires the **95th percentile** wind speed as the design basis.

The four-step workflow — executed in order:

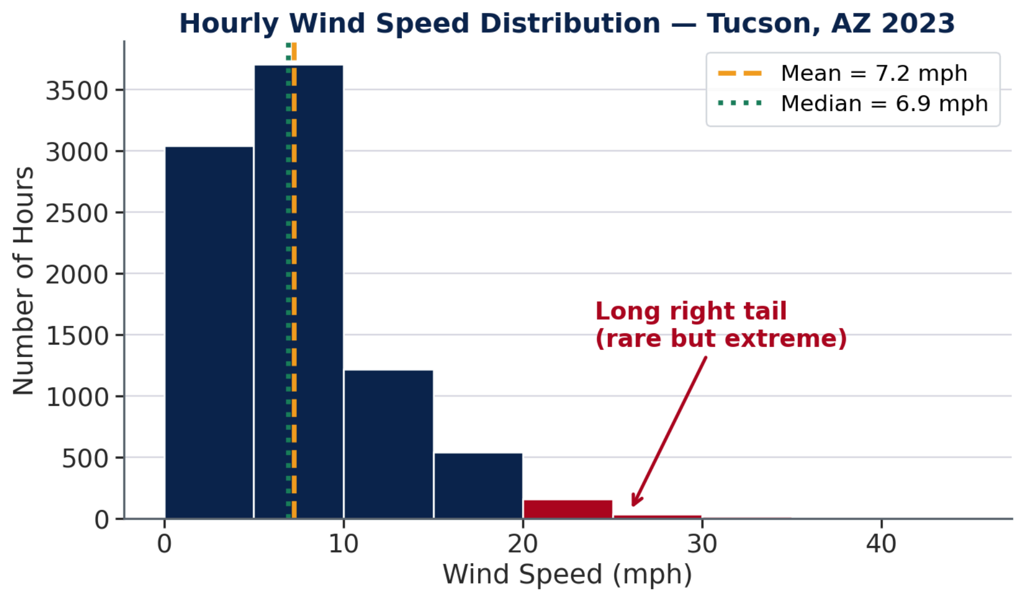
Step	Tool	Question answered
1	Histogram	What is the shape of the wind distribution?
2	Skewness coefficient	Which direction — and by how much?
3	Box plot	Where is the middle 50%? Are extreme values real?
4	Percentile	What is the actual P95 value for the design basis?

This week's lab (Sections B7–B11) puts you through exactly this sequence.

Steps 1 & 2 — Shape: Histogram and Skewness

Step 1 — Read the histogram:

- **Modal bin: 0–5 mph** — most hours are calm
- **Long right tail** extending beyond 20 mph
- Very few hours above 25 mph — but they exist
- Hard floor at zero: speed cannot be negative



Step 2 — Confirm with the coefficient:

> g = **+1.07** · ✓ positive → right-skewed · ✓ magnitude > 0.5 → meaningful asymmetry

The mean (7.2 mph) sits far below the dangerous tail.

> ⚠️ Right-skewed data: the mean underestimates the design condition. You must go to the tail.

Steps 3 & 4 — Box Plot and Percentile

Step 3 — What the box plot confirms:

Box plot feature	Value	Engineering meaning
IQR (Q1 to Q3)	4.5 mph	Middle 50% of hours span only 4.5 mph — most time is calm
Upper whisker endpoint	~17 mph	Typical non-extreme upper bound
Outlier dots	> 17 mph	Verified extreme events — real, not sensor errors

Step 4 — Find the 95th percentile:

> 8,700 × 0.95 = 8,265th-lowest value → **P95 = 17.2 mph** (342 hr above this)

Why 342 and not 8,700 – 8,265 = 435? Because 113 hours read *exactly* 17.2 mph. Subtracting ranks assumes every value is distinct; counting the record does not. 8,245 below + 113 equal + 342 above = 8,700.

> 💡 Wind pressure ∝ V^2. At P95 (17.2 mph): 17.2^2 = 296 units. At the mean (7.2 mph): 7.2^2 = 52 units. **Ratio: 5.7× — designing for average speed produces a structure 5.7× underbuilt at P95 conditions.**

The Workflow — Why Order Matters

Four tools, used in sequence, answered one engineering question:

Step	Finding	Decision it enables
Histogram	Right-skewed, long upper tail	Cannot use the mean as design basis
g = +1.07	Meaningfully asymmetric	A percentile is required — not a summary statistic
Box plot	IQR = 4.5 mph; outliers confirmed real	Extreme events are not measurement errors — include them
P95 = 17.2 mph	342 hr above threshold	Design basis confirmed; proceed to structural calculation

> 💡 The sequence matters. Skipping to step 4 without steps 1–3 means you know P95 = 17.2 mph but cannot explain *why* the mean fails.

The Texas grid engineers knew the average. What they skipped: step 1.

What Is Exploratory Data Analysis (EDA)?

> 📘 **EDA** is the practice of examining a dataset — through summaries and visualizations — before drawing conclusions or building models. Coined by statistician John Tukey (1977).

The four goals of EDA:

Goal	Question you're asking
Describe	What are the center, spread, and shape of each variable?
Discover	Are there patterns, clusters, or seasonal structure?
Check	Do the data meet the assumptions of the analysis I plan to run?
Flag	Are there outliers, gaps, or errors that could distort results?

EDA is not a formal test — it is a conversation with your data before you commit to an analysis. This week's lab puts all four goals into practice: **Describe** (A1–A9 · summary statistics), **Discover** (histogram patterns), **Check** (skewness, box plots), **Flag** (outlier dots).

Part 7

This Week's Lab

What You'll Do in the Lab — Section A

Dataset: Same Tucson 2023 file from Week 1

Section A — Histogram Basics

- A1–A2: Count hours in specific temperature bins
- A3: Count hours in a third bin, and state the bin convention you used
- A4: Measure temperature skewness
- A5–A6: Wind speed bin counts
- A7: Wind speed skewness
- A8: Humidity skewness
- A9: Coefficient of variation — carried over from Week 1

What You'll Do in the Lab — Section B

Section B — In-Depth Analysis

- B1–B3: Temperature Q1, Q3, IQR
- B4–B5: Temperature 90th and 95th percentile
- B6: Hours above the 95th percentile
- B7–B11: Wind quartiles, IQR, P95, wind pressure at P95
- B12–B13: Humidity P90, August average humidity
- B14: Probability bridge — estimate $P(80 \leq T < 90)$ from your histogram

Bridge to Machine Learning — Exploratory Data Analysis

Week 2 tool	ML equivalent	Connection
Histogram of a column	Feature distribution plot	The first look at every input before modeling
Skewness	Transformation check	Strongly skewed features usually get log-transformed
1.5 × IQR outlier rule	Anomaly detection	The same rule, run automatically on every feature
Box plots by group	Class separation check	Does this feature actually distinguish the groups?

The whole routine has a name: **exploratory data analysis (EDA).** Before fitting anything, a data scientist does exactly what you did this week — plot each feature's distribution, check the skew, flag the outliers, compare the groups. Skipping it is the most common reason a model fits its training data and then fails in the field.

Three Deliverables This Week

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 8:00 AM Tuesday, Sep 8
In-Class Exercise	Sections A and B, the charts, and the memo	Work Wed + Fri · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

The exercise spans two sessions. You start it Wednesday after the tutorial, and finish it Friday. **You are not expected to finish in class** — use the session time for the parts where having me in the room actually helps, and complete the rest on your own.

> 💡 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Tuesday, Sep 8; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Looking Ahead — Week 3 Preview

At the end of this week's lab you'll see this question (ungraded):

"In last week's lab you counted the hours where outdoor temperature exceeded 95°F. Use that count to estimate P(T > 95°F)."

> P(T > 95°F) = 900 hours / 8,760 hours = **10.3%**

The question to sit with over the week:

- Is this the **true probability** of a hot hour in Tucson?
- Or is it an **estimate** based on one year of data?
- What would you need to be more confident?

> 💡 Week 3 is about conditional probability — P(A | B) — and will build directly on the histograms and percentiles you compute this week.

Summary — Week 2

Shape is information that summary statistics discard

The mean and standard deviation describe center and spread — but two distributions with identical mean and SD can look completely different when you plot them.

Today's tools:

- **Histogram** — visualize the full distribution
- **Skewness coefficient** g — quantify asymmetry
- **Box plot** — five-number summary at a glance
- **Percentile** — locate any point in the distribution

The core principle: Design for the **tail**, not the center. The histogram and percentile are how you find the tail.